EL NIÑO ACTION BRIEFING · 25 SEPTEMBER 2026 · VERIFIED RUN

Santa Teresa, Costa Rica

The only hazard here with recorded deaths is the sea, and those beaches are already guarded. The El Niño peak lands on the tourist peak, on the same wells, one road, and no fire station; the first move is a situation report from a committee nobody has confirmed exists.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Puntarenas / Cóbano (Climate Impact Lab region containing Cóbano district)	No figure published for Costa Rica	Jun 2026 – Feb 2027 (heat only; not applicable here)	Peak-season demand on a coastal aquifer; drowning; wildfire with no station; dengue after storage

17/17 recommendations verified. 10 corrected (partially confirmed), 1 confirmed, 3 unverified, 3 contraindicated. Two lens citations struck as fabricated or misattributed. The funding target moved from $105k to $111k, under a third, so the envelope stands. **Method caveat:** the session's web-search budget was exhausted before this run; all thirteen agents worked by direct fetch of named URLs. Claims they could not reach are marked unverified rather than silently accepted. Sites unreachable this run: the fire service, the CNE, MTSS wage tables, the Health Ministry's regional health analyses, and any Cóbano council domain. Exposure figure: Climate Impact Lab, Sep 2026, pre-peer-review, publishes no Costa Rica figure. Nothing here is scaled from it.

EXECUTIVE SUMMARY — AT A GLANCE



9.64°N 85.17°W, Costa Rica · Natural Earth

- **The risk.** Peak tourist demand strains one coastal aquifer, but only the sea has killed: three drowned in one day at unguarded Montezuma and Cabuya in 2015, and a snorkeller at Cabuya in April 2026.
- **First action.** The Intendente, as Cóbano emergency committee chair, holds one session by 15 October 2026 to confirm recognition and file a situation report with well levels, copying the Puntarenas committee.
- **The money.** $30k floor, $111k target, $136k ceiling. Top placement: $25k to the Santa Teresa lifeguard association for towers at Montezuma and Cabuya, drowning being the one hazard with a death record.
- **Do not.** Do not drill or deepen a coastal well before chloride monitoring exists, or with hotel money for extraction, because Playa Hermosa lost three of nine wells and salinised the rest that way.

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* Suggested formulation: check with a health worker before sharing.

EXPOSURE, HONESTLY

WHAT IS AND IS NOT QUANTIFIED

Cóbano is a district of Puntarenas canton with its own district council; the bill to make it a canton was returned to first debate in February 2025 and is not law. Puntarenas canton has been on **orange** alert since 18 September, but no Cóbano water system appears on the national rationing list as of 24 September. That means unreported, not safe.

- **People.** INEC 2022: Cóbano district 10,490 residents, up 40% since 2011; 5,453 dwellings of which **1,956 (36%) are unoccupied**, a second-home and rental stock; 3.0 persons per occupied dwelling. The high season multiplies the population several-fold onto the same wells. Puntarenas tourism rose from 132,768 to 531,304 visitors between 2006 and 2017.
- **Rainfall.** Pacífico Norte ran 37% below normal January to August 2026 (IMN via CNE); the September to November outlook is about 30% below normal across the Pacific; El Niño is forecast "very strong" November to January. The nearest IMN automatic stations are Paquera, Puntarenas, and Aranjuez; none is in Cóbano.
- **Drowning.** The one hazard with a local death record. Segura et al. 2022: 936 coastal drowning deaths in Costa Rica 2001–2019, 38% foreigners, **70% of them in Puntarenas province**; the leading cause of unintentional death among foreign visitors. Locally: two Canadians aged 70 and 2 at Playa Hermosa, January 2015; three in one day at unguarded Montezuma and Cabuya that season; a snorkeller at Cabuya, April 2026.
- **Water.** Santa Teresa ran on tanker trucks every dry season until AyA's Bajos de Ario aqueduct opened in 2018. The community water board formed in 2010 is last documented in 2011. **Who operates the supply today could not be established from any reachable source**, and it decides three of the actions below.
- **Fire and dengue.** Cóbano has no confirmed fire station; the June 2026 Los Mangos fire that took three homes was attended from Paquera, with first water from a neighbour's cistern truck. Dengue is up 52% nationally at week 34, with the Central Pacific region the highest rate; a DENV-4 cluster was confirmed in Cóbano in November 2022.

WHAT THIS PLACE SHOULD DO

RANKED

Ranked by expected harm averted per unit of effort before the Nov–Jan peak, then by evidence quality after verification. The Pavones ranking

* Suggested formulation: check with a health worker before sharing.

does not transfer: Cóbano was inside the 2014 drought decree and received nothing on record for three years, so inclusion is not the lever here.

One session of the Cóbano emergency committee, by 15 October: confirm it is recognised, and file a situation report with the operator's well levels attached

PARTIALLY CONFIRMED | EMERGENCY MGR · ECONOMIST · HISTORIAN · PHILANTHROPIST | ADAPTATION INVENTORY: CLOSEST PRECEDENT FONDEN (MEXICO)

OWNER	BY WHEN	COST	PREVENTS
Intendente as CME chair; copy the Puntarenas CME	15 Oct 2026	Zero	Being absent from allocation when the wells fall in December

The orange alert on Puntarenas canton already unlocks CNE first-impact spending under Ley 8488 Art. 15 without a decree, and the IMAS emergency transfer runs off an incident report from a CME, Cruz Roja, or the CCSS. What is missing is the paper. Whether Cóbano's committee is recognised by the CNE in its own right could not be fetched; file through it and copy the Puntarenas committee so the request is valid whichever the CNE recognises. Moracia de Nicoya received storage containers this month through its CME, not from the CNE directly.

Why this outranks the decree petition that led the Pavones briefing: the 2014 decree's Article 1 names "Cantón Central de la provincia de Puntarenas," which includes Cóbano, and the Historian could find nothing Cóbano-specific delivered until the aqueduct opened in 2018. A district council drought resolution, as Nicoya's council passed on 21 July 2014, "increases the probability of obtaining resources" and unlocks nothing by itself; pass it as a supporting step, not the action.

Weekly static level and chloride on every operator and registered private well from 15 October; cap pumping per well; shut the first one trending saline

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · HISTORIAN · FRONTLINE · ECONOMIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
The operator (AyA Región Pacífico Central or the ASADA) + CME	First readings 15 Oct; public dashboard by 1 Dec	$5k kit; staff time	The Playa Hermosa script: 3 of 9 wells out, the rest salinised

* Suggested formulation: check with a health worker before sharing.

Five lenses converged from five directions, and the convergence survives the shared-bias check because the precedents are next door and were fetched: Playa Hermosa and Panamá lost three of nine wells fully and the other six partially to salt after 2015–16, fifteen years after UCR and AyA warned; Nosara's board drilled two unpermitted wells in 2016 beside more than 400 clandestine ones; Tamarindo lost one of its best wells to intrusion in 2015. None of it was measured until it failed. The register matters as much as the gauge: the 2008 court halt on new wells at Hermosa failed because private extraction was never counted.

Corrections: no SENARA or AyA study of the Bajos de Ario or Santa Teresa aquifer could be found, so the salinity risk is inferred from analogues, not documented locally. "Half of Guanacaste wells unregistered" is a hydrologist's "se dice"; the true number is unknown. The precondition for this whole card is one call to AyA Región Pacífico Central to establish who operates the system.

Fund lifeguard towers at Montezuma and Cabuya, and paid backfill for the Santa Teresa association's volunteer roster, 15 December to 15 April

PARTIALLY CONFIRMED · REDESIGNED | EMERGENCY MGR · PUBLIC HEALTH · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Asociación Guardavidas Santa Teresa–Malpaís–Hermosa; Concejo de Distrito under Law 9790	**Hiring by 1 Dec; towers up 15 Dec**	**~$25k (wage unverified; MTSS tables unreachable)**	**The 1–3 preventable drownings the record shows per high season**

The original ask, "daily cover at Santa Teresa, Carmen, and Hermosa," was redirected by verification. The association's own site reports 365 patrol days, 8:00 to 18:00, five towers and three stands, 216 rescues and 6,628 preventions in 2025, 85% volunteers, self-funded through its gym and some forty corporate sponsors. Those beaches are covered. **The unguarded gap is Montezuma and Cabuya**, where three people drowned in one day in 2015 and where the April 2026 death occurred. Paying salaried guards into a volunteer organisation is a change of operating model, and hotel matching for it may relabel donations the same hotels already make.

* Suggested formulation: check with a health worker before sharing.

Evidence corrections: Segura 2022 confirmed verbatim, and it notes **Law 9790 (2019) obliges coastal municipalities to provide permanent lifeguards**, delayed by COVID; ICT's registry lists a Cóbano municipal representative on the national commission but does not list this association among the seven registered organisations. Two citations were struck: the "Heliyon 2022 unpatrolled beaches" claim is actually Uebelhoer et al. 2022, 73% of Australian coastal drownings more than 1 km from a patrol or outside patrol hours, a proximity correlation and not an effect size; "Veisani 2026, 33-fold" does not exist. Do not quote a multiplier.

Lids, covers, chlorine, and brigadista-style container rounds in the worker barrios, timed to the first supply cut

PARTIALLY CONFIRMED · FRONTLINE · PUBLIC HEALTH · PHILANTHROPIST · EMERGENCY MGR; SKEPTIC: MEASURE, DO NOT ASSUME · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Waterkeeper or ADI de Mal País y Santa Teresa as distributor; Área de Salud Peninsular supervising	Stock in hand by 1 Dec; rounds from first cut	~$33/household; $46.2k for ~1,400 hh (assumption)	Diarrhoea (~25% less, low certainty) and stored-water Aedes

The Camino Verde trial (BMJ 2015, 150 clusters, Managua and coastal Guerrero) is real: dengue infection down 29.5% (3.8–55.3), house index down 44%, pupae per person down 52%. But it was community mobilisation with brigadista visits and clean-ups, not a tank-tipping protocol tied to a rationing calendar, and it ran in stable households. A town where 36% of dwellings are empty and the renters are transient is the opposite setting; the transfer is plausible and untested. Household chlorination RR 0.77 and filtration RR 0.48 (Cochrane) are confirmed, on illness not deaths. Larvicide costs nothing here: the Ministry supplies it and Costa Rican bioassays show no temephos resistance.*

* Suggested formulation: check with a health worker before sharing.

The Skeptic's counter, that the 2014 drought cut dengue 79% nationally, is a same-season observation and does not undercut Lowe 2021's finding that risk rises three to five months after drought where supply is unreliable. Both can be true. What settles it locally is a February larval survey, which is cheap. The Frontline lens's point stands: the rooms behind the hotels lose water first, and nobody reports it because the tenant is undocumented and the landlord is the employer. Count taps, not announcements.

Order storage tanks by 15 October, and sign a tanker bridge that expires on 30 April 2027, metered into an operator-owned tank, matched one-to-one by hotels through the operator's assembly

PARTIALLY CONFIRMED · ECONOMIST · EMERGENCY MGR · HISTORIAN · PHILANTHROPIST; SKEPTIC DISSENT RESOLVED · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	TRIGGER
The operator; hotels as matching payers, never as owners	Tanks ordered 15 Oct (order-only, 6–8 weeks); contract by 30 Nov	~$21k ($8k tanks + $8k standby + $5k kit; prices unverified)	Well static level below a board-set threshold, read weekly

The Economist's per-room-night escrow had no Costa Rican precedent and its room count was garbled from a national ICT table, so it survives only as a matching mechanism. The Skeptic's Sardinal warning was tested and narrowed: what detonated in 2008 was hotel-funded *extraction*, a 22-business trust for aquifer withdrawal annulled by the Sala IV; hotel money into an operator-owned tank and a time-limited bridge is a different thing. Tamarindo 2015 is the confirmed case for the expiry date: two trucks a day "while the problem is solved definitively," drivers calling in sick, hotels spending more than $1M on trucked water through 2016, residents still buying trucks in 2024 while permits kept issuing.

Rest, shade, and water protocol for construction and hospitality crews, triggered on an IMN forecast maximum, from the first hot week of December

PARTIALLY CONFIRMED · PUBLIC HEALTH · PHILANTHROPIST · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Tourism chamber + contractors; CCSS occupational health	1 Dec 2026	~$5k (water stations, shade, training)	Kidney injury in outdoor workers on a dry-Pacific coast

* Suggested formulation: check with a health worker before sharing.

Guanacaste's male age-adjusted kidney-disease mortality rose from 4.4 to 38.5 per 100,000 between 1970 and 2012 (Wesseling 2015, confirmed), on the same dry-Pacific coast. The lens's link to Nicaraguan migrants was inference, not a finding. The intervention evidence is Nicaraguan sugarcane rather than Costa Rican hospitality: under rest-shade-hydration, incident kidney injury fell from 21% to 1% and acute injury from 20 to 8 per 1,000 worker-months over four harvests (Hansson 2025), at a return of $1.60 per dollar (Schlader 2025). IMN issues no formal heat-advisory tier; its high-temperature warnings are forecast remarks, so the trigger has to be written as a forecast maximum or a wet-bulb threshold, not an alert colour.

A SINAC-accredited volunteer fire crew for Santa Teresa–Cabuya on the Nosara model, a water tender pre-positioned in Cóbano, and a MAG burn-permit hold, all by 1 February

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
SINAC Cabo Blanco + Bomberos Paquera + CME; MAG regional agency for permits	Training certified by Nov; crew and tender by 1 Feb 2027	~$4k cache (unverified); tender contract unpriced	The June 2026 pattern: three homes gone before the first unit arrives

Confirmed: the Los Mangos de Cóbano fire of 12 June 2026, three homes lost, attended by Bomberos Paquera, first water from a neighbour's loaded cistern truck; the Nosara fire of March 2019, held by an independent volunteer brigade for three to four days until official support arrived, with the nearest station ninety minutes away and El Niño drought named. Not confirmed: the February 2024

* Suggested formulation: check with a health worker before sharing.

Cóbano wildfire's size (the headline exists, the body could not be read), the Paquera station's 2001 opening and one-hour response, and an April 2026 announcement of a Cóbano station "very soon." The fire service's site is behind a captcha. Correction on the lever: MAG's regional extension agencies issue burn permits under Decreto 35368, not SINAC, and no routine dry-season blackout was found.

Post cut schedules in Spanish, English, and Hebrew through the chamber and barrio WhatsApp groups, and add a two-question tap check to Waterkeeper's river rounds

UNVERIFIED · FRONTLINE · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Nicoya Peninsula Waterkeeper (7 staff, 20 sampling sites) + volunteers	From the first cut; rounds from 1 Nov	Volunteer time	Warnings that never reach the rented rooms

Unverified because no rationing schedule exists on record to post, and the operator who would issue one is unknown. The mechanism is sound and costs nothing: Waterkeeper already samples twenty river sites monthly, ran 23 clean-ups with 404 volunteers in 2022, and sits at Plaza Kahuna in the town. Two questions at every stop, "do you have water today, who lives here," produce the first map of worker housing. The Hebrew-speaking community's size could not be checked.

Pre-position chlorine, oral rehydration salts, and fuel in Cóbano by December, and plan as if cut off for three days

PARTIALLY CONFIRMED · SKEPTIC · EMERGENCY MGR · ADAPTATION INVENTORY: PRECEDENT, NICARAGUA BRIDGES (FLOOD-CUT ACCESS)

OWNER	BY WHEN	COST	PREVENTS
CME + Cruz Roja Cóbano + EBAIS	1 Dec 2026	Minimal	Dependence on the Paquera ferry and one dirt road at peak load

* Suggested formulation: check with a health worker before sharing.

Corrected: the CNE suspended the Puntarenas–Playa Naranjo ferry from the evening of 5 October 2017 for Nate; Otto did not stop the ferries; no record shows the Cóbano road cut in either. A three-day isolation is a prudent planning assumption, not a documented precedent, and the report says so.

IMAS emergency transfer for workers laid off when supply fails, generated off the same CME incident report

PARTIALLY CONFIRMED | ECONOMIST | ADAPTATION INVENTORY: PRECEDENT, ANTICIPATORY CASH (BANGLADESH, NEPAL); CCT UNDER DROUGHT (MEXICO)

OWNER	BY WHEN	COST	PREVENTS
IMAS; CME issues the incident report	Report template ready by 1 Dec	Unpriced; the lens's ₡200k × 2 months is unsupported	Income collapse for seasonal workers if hotels close rooms

IMAS's "Emergencias" benefit exists and requires an incident report from the CNE, a CME, Bomberos, Cruz Roja, the CCSS, or the Health Ministry, plus registration and a valid identity document verified with the TSE or migration. Amount, duration, and drought-layoff eligibility are not stated, and **the ID requirement will exclude irregular-status migrants**, who are the workers most exposed. The Adaptation Inventory's strongest entries are cash: forecast-based transfers in Bangladesh and Nepal beat the same cash months later, and Mexico's PROGRESA fully offset drought's effect on fruit and vegetable intake.

EBAIS heat-day calls to elderly and chronic-disease patients

CONFIRMED EVIDENCE · WEAK TRANSFER | PUBLIC HEALTH | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CCSS Área de Salud Peninsular	Hot weeks from December	Within CCSS	Single-digit deaths at most, statistically invisible

* Suggested formulation: check with a health worker before sharing.

Hess 2018 is confirmed: Ahmedabad's heat action plan cut the mortality risk at 47 °C from 2.34 to 1.25, about 1,190 deaths a year avoided in a city of seven million, in a before-and-after design with no control. Cóbano is 10,490 people in humid coastal heat around 34 °C. The expected absolute effect is undetectable. Ranked low on that basis, kept because it costs nothing.

Interim dust suppression on the seven school road segments, December to April

UNVERIFIED · HISTORIAN · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Not identified	—	Unpriced	The dry-season asthma spike the 2013 campaign documented

The 2013 campaign (500+ children, seven named schools, a pharmacist's respiratory quote), the February 2018 suit, and the September 2024 settlement for a 218 km network with no cost, contractor, or date are all confirmed. No paving under way in 2025–26 was found, and no precedent for municipal water-truck dust control in Cóbano. An action with no owner is a wish; it is listed so someone can claim it.

Post-peak: a parametric design study with INS and IMN on a wet-season index at Paquera, settled 1 November

UNVERIFIED · NOT BEFORE 2027–28 · ECONOMIST · ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Operator + tourism chamber + INS	Study in 2027	Unpriced	Paying on noise: December is always dry

The Economist's one thing survives: Santa Teresa's drought is a recharge problem, so a dry-season rainfall trigger pays on noise; index May to October and settle on 1 November, before the tourists arrive. This year that window has closed. No insurer writes such a product, the Paquera feed is "preliminary, without quality control" and forty kilometres away on the gulf side, and the SUGESE–ICT–GIZ tourism platform launched in July 2026 is a risk-index tool with insurance "in the medium term."

* Suggested formulation: check with a health worker before sharing.

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Cóbano CME (Intendente)	Puntarenas CME	CNE	Water operator	Hotels, chamber & contractors	Guardavidas association	Concejo de Distrito	Waterkeeper, ADI & volunteers	CCSS Área de Salud Peninsular	SINAC Cabo Blanco & Bomberos Paquera	MAG regional agency	Cruz Roja Cóbano	IMAS	INS & IMN
1. Committee session and situation report	**A/R**	**I**	**I**	**C**										
2. Weekly well levels and chloride; cap pumping	**R**			**A/R**										
3. Lifeguard towers at Montezuma and Cabuya					**C**	**R**	**A**							
4. Lids, chlorine, and container rounds				**I**				**R**	**A**					
5. Storage tanks and an expiring tanker bridge				**A/R**	**R**									
6. Heat protocol for outdoor crews					**A/R**				**C**					
7. Fire crew, water tender, burn-permit hold	**R**									**A/R**	**R**			
8. Trilingual cut schedules and tap checks				**C**	**R**			**A/R**						
9. Pre-position chlorine, ORS, and fuel	**A/R**								**R**			**R**		
10. IMAS transfer for laid-off workers	**R**												**A/R**	
11. EBAIS heat-day calls to vulnerable patients									**A/R**					
12. Dust suppression on school roads							**A/R**							
13. Post-peak: parametric insurance design study				**R**	**R**									**A/R**

What the chart shows. **The Cóbano emergency committee appears on five rows (1, 2, 7, 9, and 10), and whether the CNE recognises it is unconfirmed.** That is why row 1 copies the Puntarenas committee. If the paperwork fails, the Art. 15 spending and the IMAS transfer in row 10 fail with it. **The water operator is accountable for rows 2 and 5 and appears on four more, but it has not been identified.** One call to AyA Región Pacífico Central settles it, and every water row waits on that call. **Row 12 has no named owner in the report.** The A sits with the Concejo de Distrito only because the text points to municipal roads and municipal water-truck dust control, and nobody has claimed it. **Row 3 puts the A on the Concejo de Distrito because Law 9790 obliges coastal municipalities to provide permanent lifeguards.** The association, 85% volunteers, does the work, and hotel matching may only relabel donations its sponsors already make.

* Suggested formulation: check with a health worker before sharing.

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal harm averted per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$30k	**$111k**	**$136k**	**Who operates the water, and the lifeguard placement's design**

Arithmetic. Floor: lifeguard placement $25k plus the gauging and leak kit $5k = $30k. Target: $25k + $46.2k household kits + $21k tanks and bridge + $5k heat protocol + $4k fire cache = $101.2k, plus 10% fiscal-agent overhead = $111k; it moved from $105k, up 6%, under a third, so the envelope stands. Ceiling: target plus a $15k second tanker window and $10k second tank set = $136k; the lens's "$20k all-daylight lifeguard cover" was struck because the towers already run 8:00 to 18:00, past December sunset. Household count: INEC 2022 gives 3,497 occupied dwellings; 40% as worker housing is an assumption the census cannot check, because informal migrant housing is what a census undercounts, so 1,399 × $33 = $46.2k. **Matching changes the mix, not the size:** if hotel and resident money carries the lifeguards and the tanks (~$46k), the grant need falls to about $65k with the ceiling unchanged, because the ceiling is absorptive, not financial. Caveat: the lifeguard association's sponsor base is already the hotel and restaurant community, so matching there may relabel existing donations.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Asociación Guardavidas Santa Teresa–Malpaís–Hermosa, via Amigos of Costa Rica (US fiscal sponsor)	Action 3: towers at Montezuma and Cabuya; paid backfill for the volunteer roster	$25k	Direct grant in USD; hotels matching	Only recipient with an audited-level channel (badges Finance: Tree, Governance: Seed, Impact: Sapling). No published budget or guard count. Wage ~$1,000/month loaded is plausible but unsourced; MTSS tables unreachable. Drowning is the one hazard with a death record.	Hold, redesigned

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Nicoya Peninsula Waterkeeper or ADI de Mal País y Santa Teresa de Cóbano (DINADECO code 1443, active; list header dated March 2023); Área de Salud Peninsular supervising	Action 4: lids, covers, chlorine, container rounds for ~1,400 households	$46.2k	Pre-positioned stock, direct grant	Kit figure carried from the Pavones run. Waterkeeper has 7 staff and small distribution experience (100 compost boxes, 500+ trained), nothing at 1,400 households and nothing on dengue. CEPIA does not operate here (Guanacaste only). Larvicide $0, Ministry in-kind.	Partially confirmed, adjusted up
The water operator (to be confirmed)	Actions 2 and 5: gauging and leak kit, three 10 m³ tanks, tanker bridge with expiry	$21k	1:1 matching grant with hotels through the operator's assembly	Tanks confirmed order-only, no listed price; standby unpriced. Not duplicated: no CNE, AyA, ICT, or council line for Cóbano found, but none of those sites was reachable, so this is not cleared.	Partially confirmed
Tourism chamber + contractors	Action 6: heat protocol	$5k	Direct grant	Nicaraguan sugarcane evidence, ROI $1.60 per $1; no hospitality worksite study.	Unverified cost
SINAC Cabo Blanco / Bomberos Paquera	Action 7: fire cache	$4k	In-kind	No cost-per-hectare evidence; fire service site unreachable.	Unverified

The funder's one thing, restated after verification: private money here will find the lifeguards and the hotels' own water without help. The grant dollar's only unique job is the fourteen hundred worker households that will be filling open barrels by January, and the operator's gauges, which nobody else will pay for because nobody else can see the aquifer.

* Suggested formulation: check with a health worker before sharing.

DO NOT DO THIS CONTRAINDICATED

Advice that circulates for this hazard and would do harm here. Every one of these has a confirmed local or next-door precedent.

Do not sign tanker trucking without an expiry date

CONTRAINDICATED SKEPTIC; CONFIRMED ON FOUR SOURCES

Tamarindo, August 2015: two trucks a day "while the problem is solved definitively," no end date. October 2015: trucks "not always consistent, drivers call in sick." February 2017: hotels in Carrillo and Santa Cruz spent more than $1M on trucked water across the 2015 and 2016 seasons. February 2024: Cerro Khun residents 10 to 27 days without water, buying trucks at 10 pm and 6 am, while AyA kept issuing availability letters. Trucks turn an emergency into a market and remove the pressure to fix the source.

Instead: a bridge with a 30 April 2027 sunset, a well-level trigger, and metered deliveries into an operator-owned tank, never a hotel-owned one.

Do not drill or deepen a coastal well before chloride monitoring exists, and never with hotel money for extraction

CONTRAINDICATED SKEPTIC · HISTORIAN

Playa Hermosa and Panamá: warned in 2001, new wells halted by the Sala IV in 2008, three of nine wells out and the rest salinised after 2015–16, 3,000 people on alternate-day tankers, AyA's own engineer saying coastal aquifers should never have been used. Huacas-Tamarindo and Brasilito-Potrero are salinised. Sardinal: a 22-business trust for aquifer withdrawal, its viability study annulled in 2009, police clashes in 2017, an AyA-funded aqueduct inaugurated in 2019. Hotel-funded storage is a different thing from hotel-funded extraction; keep the line where the Sala IV drew it.

* Suggested formulation: check with a health worker before sharing.

Instead: action 2. Measure every well weekly, cap pumping per well, and shut the first one trending saline.

Do not impose a blanket moratorium on new water connections

CONTRAINDICATED SKEPTIC; CONFIRMED

From September 2020 AyA withdrew 31 of Nicoya's 64 community boards' authority to issue availability letters over illegal wells and missing studies. Eighteen months later only eight had complied, because the ban removed the permit income they needed to pay for the studies. It sanctioned paperwork, not pumping, and Nosara's rationing ended when the rain came, not when the moratorium worked. Note the correction: this was AyA sanctioning the boards, not a board acting on its own; an operator wanting to pause connections petitions AyA's regional office.

Instead: metered demand caps against a public well-level dashboard. Sanction pumping, not paper.

Do not frame this as excess deaths, and do not quote a lifeguard "multiplier"

CONTRAINDICATED VERIFICATION

There is no published mortality projection for Costa Rica. Separately, two figures that circulated inside this panel do not survive: "Veisani 2026, 33-fold higher mortality at unguarded sites" does not exist in that author's work, and the "Heliyon 2022" claim is Uebelhoer et al. in a different journal reporting a proximity correlation, not an effect. The Segura paper is enough: 70% of Costa Rica's coastal drownings are in this province, and lifeguards are its explicit recommendation.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2010–11	Legal community water board formed for Santa Teresa on the José Alfaro well, AyA-approved billing	—	Last documented 2011; status today unknown

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Aug 2013	Road-dust campaign: 500+ children, seven schools, dry-season asthma spike	—	Suit filed Feb 2018
Sep–Oct 2014	Decreto 38642-MP-MAG names "Cantón Central de Puntarenas," so Cóbano is inside; August rain −60%, worst since 1950	—	Nothing Cóbano-specific on record for three years; inclusion is not allocation
Jan–May 2015	Playa Hermosa: Canadians aged 70 and 2 drown; three drown in one day at unguarded Montezuma and Cabuya; volunteer towers, weekends only, no public money	5+ in one season	First drowning-free Semana Santa at guarded beaches
Jul 2015	Solís names Cóbano, Santa Teresa, Mal País in a 12-project MINAE water programme	—	AyA awards the Bajos de Ario aqueduct bid Dec 2015
Aug 2015 – 2016	Tamarindo on two trucks a day, no end date; one well lost to salt	Hotels >$1M on trucked water	Still buying trucks in 2024
May 2016	Nosara board drills two unpermitted wells beside 400+ clandestine ones	—	AyA construction moratorium; 2022 rationing ended by rain
Oct 2016	Playa Hermosa and Panamá: 3 of 9 wells out, 6 partially saline, 3,000 on alternate-day tankers	—	"Coastal aquifers should never have been used"
Oct 2017	Tropical Storm Nate: Puntarenas–Naranjo ferry suspended; Puntarenas among hardest hit	Nationally ~570,000 without water	No Cóbano road-cut record
2018	AyA's Bajos de Ario aqueduct opens, ending dry-season tanker reliance	—	Hotels 100–400 gal/room/day; blackwater to streams; Health Ministry visits one day a week

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YEAR	EVENT	TOLL	WHAT CHANGED AFTER
Mar 2019	Nosara El Niño-season fire held by an independent volunteer brigade for three to four days	—	Station ninety minutes away
Sep 2020	AyA sanctions 31 of 64 Nicoya boards; 8 compliant after 18 months	—	Ban removed the income needed to comply
Nov 2022	Sixth DENV-4 case confirmed in Cóbano	—	—
Feb 2024	Cóbano wildfire on wind and heat, 2023–24 El Niño dry season	Size unverified	—
Sep 2024	Road-paving settlement: 218 km, 90% ballast or dirt, no cost, contractor, or date	—	No paving found under way in 2026
Nov 2025	Sámara's water board collapses; AyA takeover incomplete Aug 2026, 1,130 subscribers unbilled	"Many days" without water	The governance failure mode
2025	Santa Teresa lifeguards: 365 patrol days, 216 rescues, 6,628 preventions, 85% volunteers	—	Montezuma and Cabuya still unguarded
Apr–Jun 2026	Cabuya snorkeller death; Los Mangos fire takes three homes, attended from Paquera	1 dead; 3 homes	No Cóbano fire station confirmed
18–24 Sep 2026	Puntarenas canton orange; 32 boards rationing in 74 communities, none in Cóbano; dengue +52% nationally at week 34	—	Unreported, not safe

The recurrence nobody has noticed, from the Historian and confirmed by the record: every response here has landed after the rains. The 2014 decree produced an aqueduct in 2018; the 2018 suit produced a dateless settlement in 2024; Nosara's rationing ended by weather, not engineering. And nobody has said aloud that the demand peak, December to April, sits exactly on the recharge trough, and this El Niño's peak lands on both.

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WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Who operates the water.** The Frontline and Historian lenses describe a community board; verification shows AyA built, tendered, and commissioned the aqueduct as its own system, and the 2010 board is last seen in 2011. Actions 2, 5, and 8 and two funding placements depend on the answer. One call to AyA Región Pacífico Central settles it; no reachable source did.
- **Extend the Santa Teresa lifeguards** versus **they already cover it**. Resolved toward the latter by the association's own figures. The open question is whether it will operate at Montezuma and Cabuya at all, and whether paying salaried guards changes an organisation that runs on volunteers and a gym.
- **Dengue timing.** Public Health cites a nine-month ENSO lead pointing to August–October 2027; verification found that lead "not clearly identified" for the cluster containing Puntarenas canton. Lowe's three-to-five month post-drought lag points to spring 2027. The Skeptic's 2014 same-season decline is real and does not contradict either. A February larval survey is the cheap tiebreaker.
- **Hotel water escrow** versus **Sardinal**. Narrowed: hotel money for storage and a time-limited bridge is not what detonated at Sardinal; hotel money for extraction is. The escrow's room count was garbled from a national table and no per-room levy precedent exists in Costa Rica.
- **Guanacaste 2014–15 as the analogue.** The Skeptic argues tourism towns fail the Sámara way, by governance collapse, before the Tamarindo way, by salt. Both failure modes are confirmed next door. Which comes first here depends on the operator question above.

WHAT NOBODY COVERED — BLIND SPOT

No lens established who runs the water, and the whole briefing leans on that. No lens looked at the 1,956 unoccupied dwellings as a hazard in their own right: absent owners' pools, cisterns, and tanks are unmonitored Aedes habitat through exactly the months that matter, and no volunteer round will enter them. No lens treated tourists as recipients of warnings rather than as a load on the wells, so rip-current signage, hotel briefings, and the Hebrew and English channels for supply cuts are unaddressed. No lens costed wastewater: with no blackwater network, low stream flow concentrates what the hotels discharge, and Waterkeeper's sampling is the only monitoring. And nothing reached any Health Ministry regional analysis for the peninsula, so the age and chronic-disease structure of the resident population is unknown here.

HOW TO READ THIS — METHOD

Each briefing starts from a single, precise place, such as a village, a ward, or a district, because that is the scale at which useful actions differ. We first work out what can honestly be measured there. That usually means the Climate Impact Lab's published figures where they exist, the

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local rainfall record, census data, and the history of what has actually gone wrong in that place. Where no reliable number exists, we say so plainly rather than stretch a regional figure to fit.

We then put the same question to seven expert perspectives, each run as a separate AI research agent working from real sources. Each one asks what people here can do before the peak to keep each other alive.

- **The Emergency Manager** looks at what failed operationally in past events.
- **The Public Health Scientist** looks at who is most at risk and which interventions have measured effects.
- **The Frontline Responder** looks at how the community actually works and who people trust.
- **The Skeptic** looks for advice that has backfired in comparable places.
- **The Economist** asks who pays, who carries the risk uninsured, and whether any money is pre-arranged to arrive on a trigger.
- **The Historian** looks for the pattern that keeps repeating.
- **The Philanthropist** asks where a grant would do the most good, and how much money the place could realistically absorb before the peak.

The seven briefs are then reconciled. Agreement between lenses counts for less than it appears to, because they often draw on the same famous studies. Where one lens recommends something another calls harmful, both sides are recorded rather than quietly resolved. Every surviving action needs an owner, a date, a cost, and a named payer, or it is marked unfunded. The funding recommendation becomes a bounded range: a floor, a target, and a ceiling, each with its arithmetic shown. Finally, a separate set of agents checks every recommendation against primary sources. They test whether the evidence is real, whether it came from somewhere like this place, and whether the action can be done here in time. Anything found to be harmful is kept in the report under its own heading, because advice that would cost lives is the most important thing a reader can be warned about. The result is a strong, source-checked hypothesis from a simulated panel, not a consensus of the field, and each briefing says so.

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak by six further agents; the verdict tags carry the result. This run worked without web search: every agent fetched named pages directly, so discovery was limited to what the lenses already knew to look for, and the unverified count reflects sites that were blocked, not claims that were refuted.

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Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only* and publishes nothing for Costa Rica; drought, drowning, fire, and dengue are carried here by the historical record. The Adaptation Inventory was consulted through its data feed (96 entries, 30 countries): it holds no Costa Rica entry and nothing on coastal aquifers, lifeguards, vector control, or fire crews. Where a card says "none found," the inventory was checked and was silent.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Santa Teresa, Costa Rica. Climate Impact Lab impact region, as the report records it: Puntarenas / Cóbano (Climate Impact Lab region containing Cóbano district). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Where heat is not the dominant hazard, the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Separate verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted; the closest documented precedent for each action, or its absence, is shown in that action's tags.

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- **Health wording.** Passages where a wording error could hurt someone (covering, chlorinating, and checking stored water against dengue) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Gaps specific to this run are recorded in the verification banner and in each action's tags.

References (verification status in brackets) · Segura et al. 2022, coastal drowning in Costa Rica pmc.ncbi.nlm.nih.gov [confirmed verbatim] · Uebelhoer et al. 2022 nhess.copernicus.org [correct attribution] · Santa Teresa lifeguards santateresalifeguards.org · Amigos listing amigosofcostarica.org [confirmed] · ICT lifeguard registry ict.go.cr · ICT lodging supply 2003–2025 ict.go.cr [room count corrected] · 2015 drownings ticotimes.net · ticotimes.net · Cabuya 2026 puntarenasseoye.com [confirmed] · Decreto 38642-MP-MAG imprentanacional.go.cr [confirmed] · Cóbano canton bill observador.cr · Aqueduct remax-puravida-cr.com · ticotimes.net · Solís 2015 ticotimes.net · 2010 board malpaisbeach.com [operator today unverified] · CNE orange list observador.cr · rationing list observador.cr [confirmed] · Playa Hermosa/Panamá vozdeguanacaste.com · Nosara wells vozdeguanacaste.com · Nicoya sanctions vozdeguanacaste.com · Sámara guananoticias.com [confirmed] · Tamarindo trucking elmundo.cr · ticotimes.net · larepublica.net [confirmed] · Sardinal vozdeguanacaste.com · Hidrocec on intrusion ojoalclima.com [confirmed] · Nicoya council 2014 vozdeguanacaste.com [unlocks nothing by itself] · Camino Verde pmc.ncbi.nlm.nih.gov · García 2023 europepmc.org · Wesseling 2015 europepmc.org · Hess 2018 europepmc.org · Clasen 2015 europepmc.org [confirmed with corrections] · 2014 dengue decline ticotimes.net · dengue 2026 ticotimes.net · INEC 2022 district estimates inec.cr [confirmed, Cuadro 11 row 488] · Los Mangos fire puntarenasseoye.com · Nosara brigade ticotimes.net [confirmed] · MAG burn permits mag.go.cr · Road dust and paving ticotimes.net · ticotimes.net [confirmed] · Nate ferry crhoy.com · IMAS Emergencias imas.go.cr · IMN stations imn.ac.cr · FU-TURISMO latinoinsurance.com [confirmed] · Waterkeeper nicoyawaterkeeper.org · DINADECO list (rows 1436, 1443) hacienda.go.cr · CEPIA scope cepiacostarica.org · tanks distribuidorafama.co.cr · Unreachable this run: bomberos.go.cr (captcha), cne.go.cr (403), MTSS wage decrees (404), ministeriodesalud.go.cr regional ASIS (404), ccss.sa.cr, any Concejo de Cóbano domain, SINAC pages, Semanario Universidad (Cloudflare). Struck: "Veisani 2026" (not found in the author's works); "Heliyon 2022" (misattributed).
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (queried via wp-json/wp/v2/interventions, 96 entries; FONDEN entry; anticipatory cash entry; Nicaragua bridges entry)
Generated 25 September 2026 for Santa Teresa, Cóbano, Costa Rica. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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